Section 1: Design of Components

This section identifies the components of the platform and describes what each one does. They come from the 21 user stories in the Milestone 1 report (Section 3) and the client's clarifications (Section 4 of that report). They follow the tech stack the team chose: a React web client and one Flask API, with SQLite for development and tests and PostgreSQL for the hosted demo.

1.1 Architecture overview

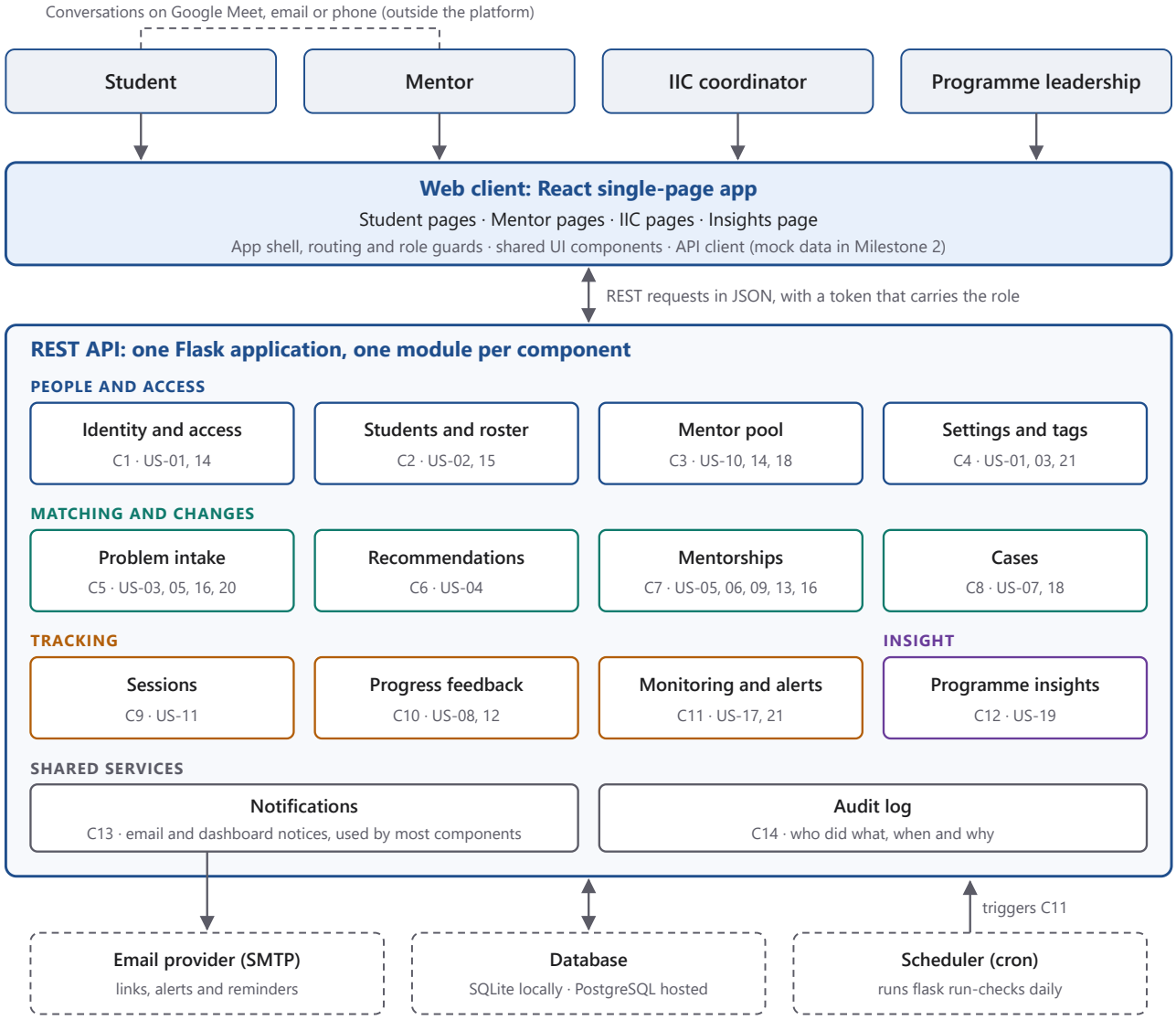


Figure 1. Components of the platform. Users reach the API only through the web client. Dashed boxes are systems outside our code, and the dashed line at the top marks conversations, which never pass through the platform. C1–C14 are described in Section 1.3.

The design follows five rules:

1. **One application, split into modules.** The API is a single Flask application with one module per component (C1–C14). Each component owns its own tables and exposes its rules through its own functions. Other components call those functions instead of writing to its tables, so each rule lives in one place.

2. **Every rule is enforced by the API.** The web client only displays data and collects input. Access control, validation, capacity checks and status changes all happen in the API, where pytest can test them.
3. **Confidential by design.** Mentoring conversations happen on Google Meet, email or phone. No component has a field for what was said (Milestone 1 report, Section 4, item 4).
4. **Predictable results.** Recommendations follow a fixed ranking rule. The timed checks take the current time as an input, so a test or a demo can run them "as of" any moment.
5. **Changeable without code.** Approved email domains, topic tags and every timing window live in one place (C4), where IIC can change them.

1.2 Components at a glance

#	Component	Main responsibility	Module	Stories	Sprint
C1	Identity and access	Sign-up, email verification, mentor activation, login and role checks	auth	US-01, US-14	1
C2	Students and roster	IIC's roster of registered students and each student's profile	students	US-02, US-15	1
C3	Mentor pool	Verified mentors, their expertise, availability and capacity	mentors	US-10, US-14, US-18	1
C4	Settings and tags	Topic-tag list, approved email domains and timing windows	settings	US-01, US-03, US-21	1–2
C5	Problem intake	Students' problems, their status, and IIC's queue	problems	US-03, US-05, US-16, US-20	1–2
C6	Recommendations	Up to 4 ranked mentors for a problem, with reasons	matching	US-04	1
C7	Mentorships	Assignment, accept or decline, goal, next steps and ending	mentorships	US-05, US-06, US-09, US-13, US-16	1–2
C8	Cases	Change, end and misconduct cases, decided by IIC	cases	US-07, US-18	2
C9	Sessions	Scheduled and logged sessions, with no content	sessions	US-11	2
C10	Progress feedback	Progress status from student and mentor, seen only by IIC	feedback	US-08, US-12	2
C11	Monitoring and alerts	IIC's monitoring view and the daily escalation checks	monitoring	US-17, US-21	2
C12	Programme insights	Aggregate demand, supply and outcome figures	insights	US-19	2
C13	Notifications	Email and dashboard notices for every event	notifications	Most stories	1–2
C14	Audit log	Permanent record of decisions and status changes	audit	US-14, US-18	1–2

The module names are the folders under backend/app in the code. Sprint 1 is Milestone 3 and Sprint 2 is Milestone 4.

1.3 Backend components

C1 Identity and access

Lets people in, and decides what each person may do. There are four roles: student, mentor, IIC coordinator and programme leadership.

Responsibilities	Student self sign-up with an institute email and roll number. Checks the email domain against C4 and the roll number against IIC's roster in C2, then sends a one-time verification link valid for 24 hours. Activates a mentor's account from the link IIC sends after verifying them (valid for 7 days). Login and logout with a token that carries the user's role. Provides the role and ownership checks that every other component uses.
Data it owns	User accounts: email, role, account status and a hashed password. Verification and activation links are signed and time-limited, so they are not stored.
Main operations	Sign up · Verify email · Activate mentor account · Log in · Log out · Get the current user
Rules it enforces	An email outside the approved domains is rejected and nothing is stored. No account is active until its link is used. One account per roll number, and no account for a roll number that is not on the roster. Calls without a valid token get 401; calls from the wrong role get 403.
Works with	C2 (roster), C3 (mentor activation), C4 (approved domains), C13 (sends the links)
Pages	Sign-up, login and account activation; the route guards in the app shell

C2 Students and roster

Holds who may join the programme (IIC's roster) and what we know about each student.

Responsibilities	IIC imports the roster from a CSV file, or adds and removes one student at a time. Each CSV row is validated: invalid rows are reported with row number and reason, and re-importing a roll number updates the record instead of duplicating it. Stores the student profile: level (BSc or BS), CGPA, completed courses and career interest. Tells C5 whether a student's profile is complete and whether they are still on the roster.
Data it owns	Roster entries (name, roll number, institute email); student profiles
Main operations	Import roster · Add or remove a roster entry · List the roster · View or update own profile
Rules it enforces	Name and roll number come from the roster and cannot be edited. Level accepts only BSc or BS, and CGPA must be between 0 and 10. A student removed from the roster cannot sign up or submit a new problem. CGPA is shown to IIC, not to mentors. Only IIC can change the roster.
Works with	C1 (sign-up check), C5 (profile complete), C14 (records imports)
Pages	IIC registered-students page; student profile page

C3 Mentor pool

Holds the verified volunteer mentors and when they can help.

Responsibilities	IIC adds mentors one at a time or from a CSV file; each starts as *Pending verification*. IIC verifies each mentor, which records who verified them and when, and triggers an activation email. Stores the mentor profile: expertise tags, experience entries (type, title and duration), mentoring style, preferred contact channel and time zone. Mentors set their availability; capacity starts at 1 and rises by 1 each time a mentor agrees to take an extra student. A mentor is removed only after an IIC decision in C8.
Data it owns	Mentor profiles, expertise tags, experience entries, verification record, availability and capacity
Main operations	Add mentor · Import mentors · Verify mentor · Resend activation link · Update own availability, channel and time zone · Accept an extra student · Remove mentor (called by C8)
Rules it enforces	A mentor is eligible for recommendation and assignment only when verified, activated, available, below capacity and not removed. Re-importing an email updates the existing profile. Unverified mentors are never visible to students. Only IIC can add, import, verify or remove mentors.
Works with	C1 (activation), C4 (topic tags), C6 and C7 (eligibility), C13, C14
Pages	IIC mentor-pool page; mentor availability and contact settings

C4 Settings and tags

Holds the values IIC can change without a code change.

Responsibilities	The topic-tag list, used both for students' problems and for mentors' expertise; the 30-tag draft list is seed data that IIC reviews before launch. The approved email domains (@ds.study.iitm.ac.in to start with). The timing windows: reminder at 48 hours and response window of 72 hours; 7 and 10 days to a first session; 14 days to *Inactive*; 7 more days to *Reassignment needed*; a feedback checkpoint every 2 weeks; *Feedback overdue* after 7 days. The default mentor capacity of 1.
Data it owns	Topic tags; setting values
Main operations	List topic tags · Add, rename or retire a tag · View or change a setting
Rules it enforces	Only IIC can make changes, and each change is recorded in C14. A changed window applies from the next run of the daily checks. A tag is retired rather than deleted, so past problems and the insights report keep their topics.
Works with	C1 (domains), C3 and C5 (tags), C10 and C11 (windows), C12 (topics), C14
Pages	Topic-tag pickers on the problem and mentor forms; IIC settings page

C5 Problem intake

Takes in each student's problem and tracks it until a mentor accepts it or IIC closes it.

Responsibilities	The problem form: a description of 30–1000 characters, 1–5 topic tags from the list in C4, an optional desired outcome and optional guidance preferences (US-20). The student can edit or withdraw it until a mentor is assigned. IIC's queue lists problems oldest first, with *Reassignment needed* problems at the top. Opening a problem sets it *Under review*. IIC can close a problem with a reason, for example when it is out of scope. The student sees the status at every step and is notified of each change.
Data it owns	Problems: description, tags, desired outcome, guidance preferences, status and timestamps

Main operations	Submit · Edit · Withdraw · View my problem's status · List the queue · Open for review · Close with a reason
Rules it enforces	The student's profile must be complete. One open problem per student, and none while the student has an active mentorship. A problem is never visible to a mentor before it is assigned to them. Guidance preferences are shown only to IIC.
Works with	C2 (profile and roster check), C4 (tags), C6 (recommendations), C7 (assignment), C13, C14
Pages	Student problem form and dashboard; IIC problem queue

C6 Recommendations

Suggests up to four suitable mentors for a problem and shows why.

Responsibilities	Takes the eligible mentors from C3 and ranks them by the number of topic tags they share with the problem (most first), then by fewest active mentees, then by mentor ID so the order never changes between runs. Returns the top 4, each with the shared tags, availability and active mentees against capacity. If no eligible mentor shares a tag, it returns an empty list and IIC browses all eligible mentors instead.
Data it owns	None. Every result is calculated from C3, C5 and C7 when it is requested, so each test case has exactly one correct answer.
Main operations	Recommend mentors for a problem · List all eligible mentors
Rules it enforces	Unverified, unavailable, at-capacity or removed mentors never appear. No machine learning. Recommendations are shown only to IIC, never to mentors or students.
Works with	C3 (eligible mentors and expertise), C5 (problem tags), C7 (active mentee counts)
Pages	IIC assignment page

C7 Mentorships

Turns IIC's choice into a working pairing and keeps its record until it ends.

Responsibilities	**Assignment:** IIC assigns a recommended or any other eligible mentor. This checks capacity, raises the mentor's active count by 1, and sends the mentor the student's level, completed courses, career interest, topic tags and problem. **Response:** the mentor accepts or declines within 72 hours; a decline needs a reason, frees capacity and returns the problem to IIC's queue. **Start:** on acceptance the mentorship becomes active and the student receives the mentor's expertise, experience, preferred channel, availability, time zone and contact details, plus an editable first-message template. The student marks "I have reached out", which records the date. **Progress:** a goal for the mentorship and next steps, each open or done with a completion date. **End:** the mentor can end the mentorship with a reason, which frees capacity; IIC can end or reassign it after a decision in C8.
Data it owns	Assignments (problem, mentor, who assigned and when, response and reason); mentorships (start date, goal, date the student reached out, status, end reason); next steps
Main operations	Assign a mentor · Accept or decline · View my mentorship · Set the goal · Add or complete a next step · Mark "I have reached out" · End the mentorship
Rules it enforces	An assignment beyond the mentor's capacity is blocked. A student has at most one active mentorship. Contact details appear only after acceptance and only to the mentee. Only the two participants and IIC can see a mentorship. The platform never sends the first message for the student.

Works with	C3 (capacity), C5 (problem status), C6 (recommendations), C8 (decisions), C9 and C10 (activity), C13, C14
Pages	IIC assignment page; mentor dashboard; mentorship workspace; mentor card on the student dashboard

C8 Cases

Handles requests to change or end a mentorship, and concerns about misconduct, fairly.

Responsibilities	A student opens a change or end request with written feedback. A student or mentor can raise a concern. While a case is open, the mentorship stays active. IIC records the other party's side, then decides: no change, reassign, end the mentorship or remove the mentor. On reassignment, the student's problem returns to IIC's queue. On removal, C3 removes the mentor and C7 ends their mentorships. Both parties are told the outcome.
Data it owns	Cases: type, who raised it, their submission, the other party's response, status, decision, reason, who decided and when
Main operations	Open a case · Record the other party's response · Decide · View the outcome of my case
Rules it enforces	Feedback in a change or end request is at least 30 characters. A concern never removes a mentor automatically. Every decision needs a reason, and the outcome, coordinator and time are recorded in C14. A removed mentor is never recommended or assigned again.
Works with	C3, C5, C7, C13, C14
Pages	Student change or end request form; IIC case page

C9 Sessions

Records when mentor and student meet, without recording what was said.

Responsibilities	Either participant schedules a session with a date and time, a channel (Google Meet, phone, email or other) and an optional meeting link. Afterwards, either marks it held, with a duration and topic category, or cancelled. A session that was not scheduled first can be logged afterwards. The other participant confirms a held session, and IIC sees both logged and confirmed counts. An optional private note is visible only to the person who wrote it. Reminders 24 hours before a session are sent by the daily checks in C11.
Data it owns	Sessions: mentorship, planned time, channel, link, status, duration, topic category, who logged it, who confirmed it, private note
Main operations	Schedule a session · Mark held or cancelled · Log a past session · Confirm a session · List a mentorship's sessions
Rules it enforces	A planned session must be in the future, and a held session cannot be dated in the future. Duration is 5–240 minutes. There is no field for conversation content. Only the participants can create or change sessions; IIC sees counts and dates, never private notes.
Works with	C7 (participants), C11 (counts, last session, reminders)
Pages	Mentorship workspace

C10 Progress feedback

Tells IIC whether a mentorship is helping, from both sides, without any ratings.

Responsibilities	The student gives a progress status (no progress, some progress or goal met) with a comment at a checkpoint or at the end. The mentor gives a progress status with an optional comment at each checkpoint, every 2 weeks by default. Records when each checkpoint falls due, so C11 can send reminders and flag *Feedback overdue*.
Data it owns	Feedback entries: mentorship, author's role, progress status, comment and time
Main operations	Submit feedback · List a mentorship's feedback
Rules it enforces	Only the three progress values are accepted, and no form has a rating. Feedback is time-stamped and visible only to IIC.
Works with	C4 (checkpoint interval), C7 (participants), C11 (reminders and flags), C12 (outcomes)
Pages	Feedback forms in the mentorship workspace; IIC monitoring and case pages

C11 Monitoring and alerts

Gives IIC one view of every mentorship and makes sure nothing stalls unnoticed.

Responsibilities	**Monitoring view:** every active mentorship with student, mentor, start date, logged and confirmed sessions, last session date, latest progress status and flags, with a filter by flag. **Daily checks**, started by the scheduler: a reminder to the mentor 48 hours after an assignment; at 72 hours without a response the assignment expires, the mentor's capacity is freed, the problem returns to the top of IIC's queue as *Reassignment needed*, and the student sees "IIC is finding you another mentor". After acceptance, reminders to both sides at 7 days without a held session and a *First session overdue* flag at 10 days. *Inactive* after 14 days without a held session, and *Reassignment needed* 7 days after that. Feedback checkpoint reminders and *Feedback overdue* after 7 days. Reminders 24 hours before each planned session.
Data it owns	Flags on assignments and mentorships (type, raised and cleared times); a record of reminders sent
Main operations	View the monitoring table · Filter by flag · Run the daily checks (by the scheduler, or with a chosen time for tests and demos)
Rules it enforces	Every window comes from C4. Each reminder is sent once, so running the checks twice changes nothing. No conversation content or private notes appear in the view.
Works with	C4, C5, C7, C9, C10, C13, C14
Pages	IIC monitoring page; alerts at the top of IIC's problem queue

C12 Programme insights

Gives programme leadership the figures to plan recruitment and judge whether the programme works.

Responsibilities	For a chosen date range: for each topic, the number of problems submitted, mentors available and problems still unassigned; the median time from problem submission to assignment; the number of active mentorships; and the outcomes of ended mentorships by final progress status. The report can be exported as CSV.
Data it owns	None. All figures are aggregates calculated from C3, C5, C7 and C10.
Main operations	Get the report for a date range · Export it as CSV
Rules it enforces	Only aggregates are shown or exported: no names or contact details. Available to the programme leadership role.

Works with	C3, C4 (topics), C5, C7, C10
Pages	Programme insights dashboard

C13 Notifications

Tells people what has happened and what they need to do.

Responsibilities	Sends an email and creates a dashboard notice for each event: verification and activation links; every change to a problem's status; a new assignment; a mentor's acceptance with their details; declines and reassignments; reminders (48 hours, first session, 24 hours before a session, feedback checkpoints); case outcomes; and a mentorship ending. Each event has one template. Emails go through the SMTP provider; in development they are printed, and in tests they are captured so tests can check them.
Data it owns	Dashboard notices: recipient, type, link, read or unread, time
Main operations	List my notices · Mark a notice as read. Other components send notifications through it.
Rules it enforces	No notification contains conversation content. A notice is shown only to its recipient.
Works with	Every component that changes something a person needs to know; the email provider
Pages	Notice list in the app shell

C14 Audit log

Keeps a permanent record of important decisions and status changes.

Responsibilities	Adds an entry (who, what, when, which record and why) for roster and mentor imports, mentor verification, assignments, declines, closures, case decisions, mentor removal, mentorship endings, settings changes and the automatic escalations from C11.
Data it owns	Audit entries, which are only ever added
Main operations	Record an entry (called by other components) · View a record's history
Rules it enforces	Entries are never edited or deleted. Entries never contain conversation content. Only IIC can view the history.
Works with	C2, C3, C4, C5, C7, C8, C11
Pages	History panel on the IIC case and monitoring pages

1.4 Web client components

The web client is a React single-page app. Pages are grouped by role, and each group only appears to users with that role.

#	Component	What it contains	Uses	Jira tasks
F1	App shell and routing	Shared layout; navigation for each role; protected routes that send users without the right role back to login; the logged-in user and role; the notice list	C1, C13	SCRUM-56
F2	Student pages	Sign-up, login and profile; the problem form; the dashboard with the status tracker and assigned-mentor card; the student's view of the mentorship workspace; the change or end request form	C1, C2, C5, C7, C8, C9, C10	SCRUM-57, SCRUM-58, SCRUM-59, SCRUM-61
F3	Mentor pages	Dashboard with assignments to accept or decline; availability, contact and capacity settings; the mentor's view of the mentorship workspace; the end-mentorship dialog	C3, C7, C9, C10	SCRUM-60, SCRUM-61
F4	IIC pages	Registered students with CSV import; the mentor pool with add, CSV import and verify; the problem queue; the assignment page with recommendations; the monitoring table; the case page; settings and topic tags	C2–C8, C11, C14	SCRUM-62, SCRUM-63
F5	Insights page	Date-range picker, demand against supply by topic, median time to assignment, active mentorships, outcomes and CSV export	C12	SCRUM-64
F6	API client	One file per backend component. In Milestone 2 it returns mock data in the same shape as the API, so pages need no changes when it switches to real requests in Milestone 3	All	SCRUM-56, SCRUM-75

Several pages share the same building blocks, which are built once:

Shared UI component	Used for
Status badge	Problem, assignment, mentorship, session and case statuses, and monitoring flags
Data table with filters	Roster, mentor pool, problem queue and monitoring table
CSV upload with row-error report	Roster and mentor imports
Reason dialog	Every action that needs a reason: decline, close, end, decide
Topic-tag picker	Problem form (1–5 tags) and mentor expertise
Form fields with validation messages	Every form; the API repeats every check

1.5 Systems outside our code

System	What it does	Notes
Database	Stores every component's data	SQLite for development and tests, so the project runs without a database server; PostgreSQL for the hosted demo
Email provider	Delivers the emails from C13 over SMTP	Needed only for the hosted demo; the provider will be chosen with hosting
Scheduler	Runs the daily checks in C11	A cron job that runs the flask run-checks command
Google Meet, email, phone	Where mentoring conversations happen	Not connected to the platform. Students and mentors use the channel the mentor prefers.

1.6 How the components work together

Getting a mentor (Sprint 1)

1. IIC loads the roster into C2 and adds mentors to C3, then verifies each mentor. C13 emails each mentor an activation link, and C14 records each verification.
2. A student signs up. C1 checks the email domain with C4 and the roll number with C2, then sends a verification link through C13. The student completes their profile in C2.
3. The student submits a problem in C5. It is saved as *Submitted* and appears in IIC's queue.
4. IIC opens the problem, which sets it *Under review*. C6 shows up to four ranked mentors with their shared tags and current load.
5. IIC assigns a mentor in C7. The capacity check passes, the problem becomes *Awaiting mentor*, and C13 notifies the mentor.
6. The mentor accepts in C7. The problem becomes *Mentor assigned*, and the student receives the mentor's details and a first-message template. If the mentor declines, the problem goes back to IIC's queue.
7. The student contacts the mentor on the mentor's preferred channel, outside the platform, and marks "I have reached out".

Tracking a mentorship (Sprint 2)

1. Student and mentor record a goal and next steps in C7.
2. They schedule and log sessions in C9. C11 sends reminders through C13 24 hours before each session.
3. At each checkpoint, both give progress feedback in C10.
4. Each week, IIC reviews the monitoring view in C11, which shows session counts, the latest progress and any flags.

When something stalls (Sprint 2)

1. Every day, the scheduler starts the checks in C11, which read the windows from C4.
2. An assignment with no response after 48 hours triggers a reminder. After 72 hours, C7 expires the assignment and frees the mentor's capacity, and C5 moves the problem to the top of IIC's queue as *Reassignment needed*.
3. A mentorship with no held session for 14 days is flagged *Inactive*; 7 days later it becomes *Reassignment needed*.
4. Every escalation is recorded in C14, and the people involved are told through C13.

Changing or ending a mentorship (Sprint 2)

1. A student asks to change or end the mentorship, or someone raises a concern. C8 opens a case, and the mentorship stays active.
2. IIC records the other side and decides, with a reason. C14 records the decision.
3. If the decision is to reassign, the problem returns to IIC's queue in C5. If it is to remove the mentor, C3 removes them and C7 ends their mentorships. Both parties are notified through C13.
4. A mentor can also end a mentorship directly in C7, giving a reason. This frees their capacity, and if they are no longer available the student is listed for reassignment.

1.7 Statuses each component owns

Each status belongs to one component, and only that component changes it.

Record	Owner	Statuses
Mentor	C3	*Pending verification* → *Verified* (activation link sent) → *Active*; or *Removed*. Availability is a separate setting.
Problem	C5	*Submitted*, *Under review*, *Awaiting mentor*, *Mentor assigned*, *Reassignment needed*, *Closed by IIC* (with a reason), *Withdrawn*
Assignment	C7	*Awaiting response* → *Accepted*, *Declined* or *Expired* (after 72 hours)
Mentorship	C7	*Active* → *Ended by mentor* or *Ended by IIC*
Session	C9	*Planned* → *Held* or *Cancelled*; a held session can be confirmed by the other participant
Case	C8	*Under review* → *Decided*
Flags	C11	*First session overdue*, *Inactive*, *Feedback overdue*, *Reassignment needed*

1.8 Traceability: user stories to components

Every user story is served by at least one component, and every component serves at least one story.

Story	Main component	Also uses
US-01 Sign up, checked against IIC's roster	C1	C2, C4, C13
US-02 Complete my basic profile	C2	C1
US-03 Submit the problem I need help with	C5	C2, C4, C13
US-04 See 3–4 recommended mentors	C6	C3, C5, C7
US-05 Track my request and reach out to my mentor	C5	C7, C13
US-06 Keep track of our goal and next steps	C7	C9
US-07 Ask to change or end my mentorship	C8	C5, C7, C13
US-08 Give feedback on my mentorship	C10	C7
US-09 Accept or decline assigned students	C7	C11, C13
US-10 Set availability, contact preference and capacity	C3	C6, C7
US-11 Schedule and log sessions	C9	C11, C13
US-12 Send progress feedback on my mentee	C10	C4, C11, C13
US-13 End a mentorship from my side	C7	C3, C13, C14
US-14 Add and verify mentors	C3	C1, C13, C14
US-15 Keep the roster up to date	C2	C14
US-16 Assign a mentor to each problem	C7	C5, C6, C13, C14
US-17 Monitor active mentorships	C11	C9, C10

Story	Main component	Also uses
US-18 Resolve change, termination and misconduct cases	C8	C3, C7, C14
US-19 See demand, supply and outcomes	C12	C3, C5, C7, C10
US-20 Say how I like to be guided	C5	C3, C6
US-21 Stop assignments and mentorships from stalling	C11	C4, C5, C7, C13, C14